

5(a)(i)	correct lines ;;	2												
5(a)(ii)	<u>basement membrane</u> ;	1												
5(b)	<table border="1"> <thead> <tr> <th>afferent arteriole lumen diameter</th><th>efferent arteriole lumen diameter</th><th>GFR</th></tr> </thead> <tbody> <tr> <td>normal</td><td>normal</td><td>normal</td></tr> <tr> <td>decreased</td><td>normal</td><td>decrease ;</td></tr> <tr> <td>normal</td><td>increased</td><td>decrease ;</td></tr> </tbody> </table>	afferent arteriole lumen diameter	efferent arteriole lumen diameter	GFR	normal	normal	normal	decreased	normal	decrease ;	normal	increased	decrease ;	2
afferent arteriole lumen diameter	efferent arteriole lumen diameter	GFR												
normal	normal	normal												
decreased	normal	decrease ;												
normal	increased	decrease ;												
5(c)	<i>any five from:</i> 1 active transport / pumping, of Na⁺ (out of cells) into, blood / tissue fluid ; 2 sodium (potassium) pumps in basal membrane ; 3 Na⁺ concentration decreases inside cells / Na⁺ concentration gradient set up ; 4 (so) Na⁺ enters (epithelial / tubule) cells from, lumen / filtrate ; 5 by facilitated diffusion ; 6 Na⁺ co-transporters glucose into cell ; 7 this is, secondary / indirect, active transport ; 8 (facilitated) diffusion of glucose (out of cells) into blood ; 9 GLUT proteins ;	5												

Question	Answer	Marks
5(d)	any three from: 1 <u>microvilli</u>, has large surface area / for increased (re)absorption ; I increase surface to volume ratio 2 many mitochondria provide <u>ATP</u> for, sodium (potassium) pumps / active transport ; 3 tight junctions / closely packed cells, hold adjacent cells together / prevents leakage / fluid cannot pass between cells / substances must pass through cells ; 4 many transport proteins for movement of named substance ;	3

Question	Answer	Marks
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